

The Garland–Lepowsky theorem in the chiral setting: bar cohomology and Koszul duality for affine Kac–Moody algebras

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Abstract

The classical Garland–Lepowsky theorem computes the Lie algebra cohomology $H^p(\mathfrak{n}_-; \mathbb{C})$ of the negative nilpotent subalgebra of a Kac–Moody algebra in terms of Weyl group combinatorics. We prove a chiral extension: for the affine Kac–Moody vertex algebra $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$, the PBW spectral sequence is controlled by Garland–Lepowsky affine nilradical homology. For $\mathfrak{g} = \mathfrak{sl}_2$ the verified Koszul-dual dimension sequence is $\dim H^n = 2n + 1$; more generally, polynomial growth is governed by the Weyl group of $\mathfrak{g}$. This diagonal PBW control is equivalent to chiral Koszulness of $V_k(\mathfrak{g})$ via the PBW spectral sequence. The Koszul dual is the chiral Chevalley–Eilenberg algebra at the Verdier-reflected level $k' = -k - 2h^\vee$, with modular characteristic $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$ satisfying the Verdier complementarity on the affine Kac–Moody line: $\kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}_{-k-2h^\vee}) = 0$. This scalar complementarity is distinct from the central-charge conductor $K_c(V_k(\mathfrak{g})) = 2 \dim \mathfrak{g}$. The shadow obstruction tower is class L with depth $r_{\max} = 3$: a nonzero cubic shadow from the Lie bracket, killed at the quartic level by the Jacobi identity.

Remark 0.1 (Bar–cobar inputs at admissible level). Two structural results from [9] are used. First, the periodic-CDG admissible-KL bar–cobar adjunction

$$\Omega^{\text{ch}} \dashv B^{\text{ch}} : KL_k^{\text{adm}} \rightleftarrows (KL_{k'}^{\text{adm}})^{\text{op}}$$

holds for the universal admissible non-critical PBW ambient at $k + h^\vee = p/q$, using Arakawa C_2 -cofiniteness and finite length. Simple quotients require the separate admissible quotient analysis. Second, the weight-completed chiral Positselski theorem gives the conilpotent counit isomorphism on the completed bar coalgebra and the co-contraction equivalence

$$D_{\text{ch}}^{\text{co}} \simeq D_{\text{ch}}^{\text{ctr}}.$$

Thus Garland–Lepowsky concentration controls the generic level and extends, through these structural inputs, to the admissible non-critical locus. At $k = -h^\vee$ the Feigin–Frenkel centre changes the bar complex; see §8.

I Introduction

The bar complex of a vertex algebra encodes the full $\mathcal{A}_\infty$ -structure of the chiral algebra in a single differential graded coalgebra. For the universal affine vertex algebra $V_k(\mathfrak{g})$, the bar cohomology $H^*(\bar{B}(V_k(\mathfrak{g})))$

measures how far the chiral algebra deviates from a free field theory. The fundamental question is: in which bar degrees does this cohomology concentrate?

For the finite-dimensional nilpotent radical $\mathfrak{n}_-$ of a Borel subalgebra of a semisimple Lie algebra $\mathfrak{g}$, Kostant [7] proved that the Lie algebra cohomology $H^p(\mathfrak{n}_-; \mathbb{C})$ is concentrated in a single Weyl group orbit at each cohomological degree p , with dimension equal to the number of Weyl group elements of length p . Garland and Lepowsky [6] extended this to the infinite-dimensional loop algebra $\mathfrak{n}_-^{\text{loop}} = \mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}]$, where the cohomology acquires a grading by conformal weight. Both results are statements about the Chevalley–Eilenberg complex. The chiral bar complex, by contrast, carries additional structure: the full singular OPE, not merely the Lie bracket. A priori, there is no reason for the classical concentration to survive in the chiral setting.

At non-critical level $k \neq -h^\vee$, the chiral bar complex $\bar{B}(V_k(\mathfrak{g}))$ has bar cohomology on the PBW Koszul diagonal. The PBW spectral sequence has E_1 -page the Chevalley–Eilenberg complex of the negative loop algebra with trivial coefficients, and the Garland–Lepowsky concentration forces $E_2 = E_\infty$.

2 The classical Garland–Lepowsky theorem

2.1 Kostant’s theorem for finite-dimensional $\mathfrak{g}$

Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, root system Δ , positive roots Δ_+ , Weyl group W , and Borel subalgebra $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+$ where $\mathfrak{n}_+ = \bigoplus_{\alpha \in \Delta_+} \mathfrak{g}_\alpha$. Let $\mathfrak{n}_- = \bigoplus_{\alpha \in \Delta_+} \mathfrak{g}_{-\alpha}$ be the opposite nilpotent subalgebra, and let ρ denote the Weyl vector (half the sum of positive roots).

Theorem 2.1 (Kostant [7]). *For each $p \geq 0$, the Lie algebra cohomology $H^p(\mathfrak{n}_-; \mathbb{C})$ is a semisimple $\mathfrak{h}$ -module:*

$$H^p(\mathfrak{n}_-; \mathbb{C}) \cong \bigoplus_{\substack{w \in W \\ \ell(w) = p}} \mathbb{C}_{w \cdot \rho},$$

where $w \cdot \rho = w(\rho) - \rho$ is the dot action and $\ell(w)$ is the length of w . In particular, $\dim H^p(\mathfrak{n}_-; \mathbb{C}) = |\{w \in W : \ell(w) = p\}|$.

For $\mathfrak{g} = \mathfrak{sl}_2$: $W = \{1, s\}$, $\ell(1) = 0$, $\ell(s) = 1$, so $\dim H^0 = 1$, $\dim H^1 = 1$, $H^p = 0$ for $p \geq 2$. For $\mathfrak{g} = \mathfrak{sl}_3$: $|W| = 6$, with length distribution $(1, 2, 2, 1)$, giving $\dim H^p = (1, 2, 2, 1)$ for $p = 0, 1, 2, 3$.

More generally, for a highest-weight module V_λ with highest weight λ :

Theorem 2.2 (Kostant [7]). *$H^p(\mathfrak{n}_-; V_\lambda) \cong \bigoplus_{\ell(w)=p} V_{w \cdot \lambda}^{\mathfrak{h}}$, where the sum runs over $w \in W$ with $\ell(w) = p$ such that $w \cdot \lambda$ is dominant.*

The proof uses the Hodge decomposition for the Laplacian $\square = dd^* + d^*d$ on $\Lambda^*(\mathfrak{n}_-)$, where the adjoint d^* is defined using the Killing form. The key input is that $\mathfrak{g}$ is semisimple: the Casimir operator provides the Hodge-theoretic splitting.

2.2 Extension to Kac–Moody algebras

Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, $\widehat{\mathfrak{g}} = \mathfrak{g} \otimes \mathbb{C}((t)) \oplus \mathbb{C}K$ the associated affine Kac–Moody algebra, and

$$\mathfrak{n}_-^{\text{loop}} = \mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}]$$

the negative part of the loop algebra (the subalgebra spanned by J_{-m}^a for $m \geq 1$ and $a = 1, \dots, \dim \mathfrak{g}$). This is a graded Lie algebra with the conformal weight grading $|J_{-m}^a| = m$.

Theorem 2.3 (Garland–Lepowsky [6]). *Let*

$$\mathfrak{u}^- = \mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}]$$

be the nilradical opposite to the standard affine parabolic, and let W_{aff}° be the minimal representatives for $W_{\text{fin}} \setminus W_{\text{aff}}$. Then

$$H_p(\mathfrak{u}^-; \mathbb{C}) \cong \bigoplus_{\substack{w \in W_{\text{aff}}^\circ \\ \ell(w)=p}} L_{\mathfrak{g}}\left(\overline{w(\widehat{\rho}) - \widehat{\rho}}\right) \otimes \mathbb{C}_{\deg_{\widehat{\rho}}(w)}.$$

Dualizing gives the corresponding cohomology statement for $H^p(\mathfrak{u}^{-\text{loop}}; \mathbb{C})$ with the opposite weight convention.

Remark 2.4. Garland–Lepowsky computes affine nilradical homology independently of Whitehead’s finite-dimensional lemma. Under the PBW filtration, the associated graded chiral bar complex is the Chevalley–Eilenberg complex of $\mathfrak{u}^-$. The PBW comparison identifies the E_2 -page with the Garland–Lepowsky homology decomposition.

3 The chiral bar complex of $V_k(\mathfrak{g})$

3.1 The universal affine vertex algebra

Let $\mathfrak{g}$ be a simple Lie algebra with invariant bilinear form $(\cdot, \cdot)$ normalized so that the highest root θ satisfies $(\theta, \theta) = 2$, and let $h^\vee$ denote the dual Coxeter number. The universal affine vertex algebra $V_k(\mathfrak{g})$ at level k is freely strongly generated by the currents $\{J^a(z)\}_{a=1}^{\dim \mathfrak{g}}$ with OPE

$$J^a(z)J^b(w) \sim \frac{k(J^a, J^b)}{(z-w)^2} + \frac{f^{ab}_c J^c(w)}{z-w}, \quad (3.1)$$

where f^{ab}_c are the structure constants of $\mathfrak{g}$ in the orthonormal basis. Free strong generation means the normally ordered monomials $:\partial^{n_1} J^{a_1} \dots \partial^{n_r} J^{a_r}: \text{ form a PBW basis of } V_k(\mathfrak{g})$.

3.2 Construction of the bar complex

The ordered bar complex of $V_k(\mathfrak{g})$ is the cofree tensor coalgebra

$$B^{\text{ord}}(V_k(\mathfrak{g})) = T^c(s^{-1}\overline{V}_k(\mathfrak{g})) = \bigoplus_{n \geq 0} (s^{-1}\overline{V}_k(\mathfrak{g}))^{\otimes n},$$

where $\overline{V}_k(\mathfrak{g}) = V_k(\mathfrak{g})/\mathbb{C}[o]$ is the augmentation ideal and s^{-1} denotes desuspension (lowering cohomological degree by 1). The bar differential d_B is the unique coderivation whose Taylor components $d_B^{(n)}: (s^{-1}\overline{V})^{\otimes n} \rightarrow s^{-1}\overline{V}$ encode the chiral algebra operations: the binary component $d_B^{(2)}$ extracts the OPE via the logarithmic kernel $d \log(z_1 - z_2)$, and the nullary component $d_B^{(0)}$ records the curvature.

Geometrically, the degree- n piece of the bar complex is

$$\bar{B}^n(V_k(\mathfrak{g})) = \Gamma(\overline{C}_{n+1}(X), \mathfrak{g}^{\boxtimes(n+1)} \otimes \omega_X^{\boxtimes(n+1)} \otimes \Omega_{\log}^n),$$

where $\overline{C}_{n+1}(X)$ is the Fulton–MacPherson compactification [5] of the configuration space of $n+1$ points on a smooth curve X , and $\Omega_{\log}^n$ is the sheaf of logarithmic n -forms along the boundary divisors.

Proposition 3.1. *The bar differential on degree-1 elements $J^a \boxtimes J^b \cdot \eta_{12}$, where $\eta_{12} = d \log(z_1 - z_2)$, is:*

$$d_B(J^a \boxtimes J^b \cdot \eta_{12}) = k(J^a, J^b) \cdot |\circ\rangle + f^{ab}{}_c J^c. \quad (3.2)$$

The first term arises from the double pole in (3.1) (the vertex product $J_{(1)}^a J_{(1)}^b$); the second from the simple pole (the product $J_{(0)}^a J_{(0)}^b$).

Proof. The residue extraction is governed by the vertex products in the OPE

$$J^a(z_1)J^b(z_2) = \frac{k(J^a, J^b)}{(z_1 - z_2)^2} + \frac{f^{ab}{}_c J^c(z_2)}{z_1 - z_2} + \text{regular}.$$

The product $J_{(1)}^a J_{(1)}^b = k(J^a, J^b)\mathbf{1}$ contributes the curvature term, while $J_{(0)}^a J_{(0)}^b = f^{ab}{}_c J^c$ contributes the collision bracket. The logarithmic form records the collision divisor; it does not identify the curvature channel with the bracket channel. $\square$

3.3 Curvature of the bar complex

Proposition 3.2. *The bar differential satisfies $d_B^2 = 0$ as a coderivation on $T^c(s^{-1}\overline{V})$. The A_∞ -curvature m_0 measures the failure of $m_1^2 = 0$ on the generating space:*

$$m_1^2 = \frac{k + h^\vee}{2h^\vee} \cdot \kappa_{\mathfrak{g}},$$

where $\kappa_{\mathfrak{g}} = \sum_a J^a \otimes J_a$ is the Casimir element. The factor $h^\vee$ arises from the identity $\sum_{c,d} f^{ac}{}_d f^{bc}{}_d = 2h^\vee \cdot \delta^{ab}$ (the Killing form in the orthonormal basis).

Proof. Acting with d_B twice on $J^a \boxtimes J^b \cdot \eta_{12}$ and applying (3.2):

$$\begin{aligned} d_B^2(J^a \boxtimes J^b \cdot \eta_{12}) &= d_B(k(J^a, J^b) \cdot |\circ\rangle + f^{ab}{}_c J^c) \\ &= f^{ab}{}_c \cdot d_B(J^c) \\ &= f^{ab}{}_c \cdot (\text{terms involving } d_B^{(2)}(J^c \boxtimes \cdot)). \end{aligned}$$

The Jacobi identity $f^{ab}{}_c f^{cd}{}_e + \text{cyclic} = 0$ combines with the Killing form computation $\sum_{c,d} f^{ac}{}_d f^{bc}{}_d = 2h^\vee \delta^{ab}$ to give the claimed factor $(k + h^\vee)/(2h^\vee)$. At the critical level $k = -h^\vee$, the curvature vanishes and $m_1^2 = 0$. $\square$

3.4 Identification with Chevalley–Eilenberg cohomology

The PBW filtration on $V_k(\mathfrak{g})$ induces a filtration on the bar complex. The associated graded is the symmetric algebra $\text{Sym}(\mathfrak{g}[t^{-1}]t^{-1})$, and the E_1 page of the resulting spectral sequence is the Chevalley–Eilenberg complex:

Proposition 3.3. *The E_1 page of the PBW spectral sequence on $\tilde{B}(V_k(\mathfrak{g}))$ is isomorphic to the Chevalley–Eilenberg cochain complex $C_{\text{CE}}^*(\mathfrak{n}_{-}^{\text{loop}}; \mathbb{C})$ of the negative loop algebra $\mathfrak{n}_{-}^{\text{loop}} = \mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}]$ with trivial coefficients.*

Proof. The PBW filtration $F_p(V_k(\mathfrak{g}))$ is defined by the total number of generators in a normally ordered monomial. On the associated graded, the OPE reduces to the commutative product (the singular part is lower-order in the filtration), and the bar differential reduces to the Lie bracket component f^{ab}_c acting on the exterior algebra $\Lambda^*(\mathfrak{g}[t^{-1}]t^{-1})^*$. This is precisely the Chevalley–Eilenberg differential. $\square$

4 Chiral Koszulness

Definition 4.1. A chiral algebra $\mathcal{A}$ is *chirally Koszul* if the bar cohomology $H^*(\tilde{B}(\mathcal{A}))$ lies on the Koszul diagonal determined by the PBW/conformal grading:

$$H^n(\tilde{B}(\mathcal{A}))_h \neq 0 \implies h = h(n)$$

for the diagonal function fixed by the PBW grading of the family. Equivalently, the PBW spectral sequence collapses at E_2 (see [9], Theorem 5.1.1).

Theorem 4.2 (Chiral Koszulness of $V_k(\mathfrak{g})$). *For any simple Lie algebra $\mathfrak{g}$ and non-critical level $k \neq -h^\vee$, the universal affine vertex algebra $V_k(\mathfrak{g})$ is chirally Koszul.*

Proof. The PBW filtration on $V_k(\mathfrak{g})$ gives a bar spectral sequence whose E_1 -page is the Chevalley–Eilenberg complex

$$C_{\text{CE}}^*(\mathfrak{n}_-^{\text{loop}}; \mathbb{C}).$$

Garland–Lepowsky compute the E_2 -page by affine Weyl representatives. Their calculation supplies the diagonal bidegree support needed for collapse; no finite-dimensional Whitehead-lemma substitute proves this affine nilradical statement.

The same conclusion is recovered from PBW generation. The universal affine vertex algebra is freely strongly generated by the currents $\{J^a\}_{a=1}^{\dim \mathfrak{g}}$, and its associated graded chiral algebra is $\text{Sym}^{\text{ch}}(\mathfrak{g}_k)$. The symmetric algebra is Koszul by Priddy [12]; the PBW universality criterion (Proposition 5.3.1 of [9]) therefore identifies the filtered chiral bar complex with its Koszul associated graded and gives chiral Koszulness. $\square$

Remark 4.3. The theorem admits two independent proofs. The first, given above, uses the Garland–Lepowsky concentration to force spectral sequence collapse. The second uses the abstract PBW universality criterion, which requires only free strong generation and Koszulness of the associated graded. That both proofs yield the same conclusion is a consistency check.

Remark 4.4 (Chiral bar cohomology vs. Chevalley–Eilenberg cohomology). The PBW spectral sequence identifies the E_1 page with the Chevalley–Eilenberg complex $C_{\text{CE}}^*(\mathfrak{n}_-^{\text{loop}}; \mathbb{C})$ and the chiral Koszulness theorem asserts that $E_2 = E_\infty$. It does *not* follow that $H^n(\tilde{B}(V_k(\mathfrak{g}))) = H_{\text{CE}}^n(\mathfrak{n}_-^{\text{loop}}; \mathbb{C})$ for all n . The chiral bar complex carries an Orlik–Solomon form factor absent from the exterior algebra $\Lambda^*(\mathfrak{n}_-^{\text{loop}})^*$: the degree- n bar group is $\mathfrak{g}^{\otimes n} \otimes \Omega_{\log}^{n-1}$, with $\dim \Omega_{\log}^{n-1} = (n-1)!$, whereas the Chevalley–Eilenberg cochain group is $\Lambda^n(\mathfrak{n}_-^{\text{loop}})^*$. For $\mathfrak{g} = \mathfrak{sl}_2$ (rank 1), the two complexes coincide at each conformal weight and the PBW collapse gives $H^n(\tilde{B}) = H_{\text{CE}}^n$. For $\mathfrak{g}$ of rank ≥ 2 , the dimensions can differ: for $\mathfrak{sl}_3$, $\dim H^2(\tilde{B}) = 36$ while $\dim H_{\text{CE}}^2 = 20$. The Koszulness conclusion (concentration in bar degree 1) is unaffected; what changes is the precise dimension count.

4.1 The affine nilradical mechanism in detail

The PBW spectral sequence collapse in the proof of Theorem 4.2 is the mechanism by which the classical Garland–Lepowsky theorem enters the chiral bar complex.

The negative loop algebra $\mathfrak{n}_-^{\text{loop}} = \mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}]$ admits a conformal weight decomposition $\mathfrak{n}_-^{\text{loop}} = \bigoplus_{m \geq 1} \mathfrak{g}_{-m}$, where $\mathfrak{g}_{-m} = \mathfrak{g} \otimes t^{-m} \cong \mathfrak{g}$ as a $\mathfrak{g}$ -module under the adjoint action. The Chevalley–Eilenberg complex decomposes accordingly:

$$C_{\text{CE}}^p(\mathfrak{n}_-^{\text{loop}}; \mathbb{C}) = \bigoplus_h \Lambda^p(\mathfrak{n}_-^{\text{loop}})_h^*,$$

where the subscript h denotes conformal weight. At each weight h , the space $\Lambda^p(\mathfrak{n}_-^{\text{loop}})_h^*$ is finite-dimensional (finitely many partitions of h into parts from $\{1, 2, 3, \dots\}$, each contributing a $\mathfrak{g}$ -factor).

The PBW spectral sequence has E_1 page

$$E_1^{p,q} = H^p(\mathfrak{g}; S^q(\mathfrak{n}_-^{\text{loop}}/\mathfrak{g})^*),$$

where S^q denotes the q -th symmetric power and the quotient $\mathfrak{n}_-^{\text{loop}}/\mathfrak{g}$ is taken modulo the weight-1 stratum. The finite-dimensional Whitehead lemma is not the Garland–Lepowsky calculation. It only controls finite-dimensional $\mathfrak{g}$ -module pieces. The required affine nilradical homology is imported from Theorem 2.3. In the finite-dimensional pieces, for a semisimple Lie algebra $\mathfrak{g}$ and a nontrivial finite-dimensional simple $\mathfrak{g}$ -module $\mathcal{M}$,

$$H^p(\mathfrak{g}; \mathcal{M}) = 0 \quad \text{for all } p \geq 0.$$

The spectral sequence comparison therefore uses Garland–Lepowsky for the affine nilradical page; it is not a Whitehead-lemma substitute.

The surviving terms at $p = 0$ are the $\mathfrak{g}$ -invariants:

$$E_1^{0,q} = S^q(\mathfrak{n}_-^{\text{loop}})^{\mathfrak{g}} = S^q\left(\bigoplus_{m \geq 1} \mathfrak{g}_{-m}\right)^{\mathfrak{g}}.$$

For $\mathfrak{g} = \mathfrak{sl}_N$, the ring of invariants of the adjoint representation under the adjoint action is generated by Casimir elements of degrees $2, 3, \dots, N$ (the exponents of the Weyl group shifted by one). At each mode t^{-m} , one obtains $N - 1$ independent Casimirs. This is the Kostant concentration in the loop setting.

Lemma 4.5. *The Euler series of the bar complex of $V_k(\mathfrak{g})$ factorizes:*

$$\chi(\bar{B}(V_k(\mathfrak{g})); q) = \prod_{n \geq 1} (1 - q^n)^{\dim \mathfrak{g}} = \eta(q)^{\dim \mathfrak{g}} / q^{\dim \mathfrak{g}/24}, \quad (4.1)$$

where $\eta(q) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ is the Dedekind eta function.

Proof. Each negative mode J_{-m}^a ($a = 1, \dots, \dim \mathfrak{g}$, $m \geq 1$) contributes a factor of $(1 - q^m)$ to the Euler characteristic, from the alternating sum of exterior powers. The product over all generators gives $\prod_{a,m} (1 - q^m) = \prod_{m \geq 1} (1 - q^m)^{\dim \mathfrak{g}}$. $\square$

Remark 4.6. The Euler characteristic (4.1) depends only on $\dim \mathfrak{g}$, not on the Lie algebra structure. For two Lie algebras with the same dimension (e.g., $\mathfrak{sl}_2$ and the three-dimensional abelian Lie algebra $\mathbb{C}^3$), the Euler series agrees: both give $\prod (1 - q^n)^3$. However, the individual cohomology dimensions $\dim H^p$ at each conformal weight differ: the Garland–Lepowsky concentration forces concentration into a single degree for each weight when $\mathfrak{g}$ is semisimple, but the actual degree depends on the root system. This distinction is emphasized in the monograph [9]: Euler characteristic = dimension only when concentration holds, which requires semisimplicity.

4.2 Bar cohomology dimensions

Theorem 4.7 (Garland–Lepowsky nilradical concentration). *Let $\widehat{\mathfrak{g}}_k$ be the affine Kac–Moody algebra associated to a simply-laced simple Lie algebra $\mathfrak{g}$ of type ADE and rank $r = \text{rk}(\mathfrak{g})$, at generic (non-critical) level $k \neq -h^\vee$. Then the Lie-algebra cohomology $H^*(\widehat{\mathfrak{g}}_k^-; \mathbb{C})$ of the negative part $\widehat{\mathfrak{g}}_k^- = \mathfrak{n}_-^{\text{loop}}$ is concentrated in a single cohomological degree at each conformal weight. The affine Weyl decomposition of Theorem 2.3 gives the E_2 -page of the PBW spectral sequence for the chiral bar complex. It does not by itself give a closed Hilbert series for the full chiral bar cohomology in every rank.*

Proof. Garland–Lepowsky [6] compute the affine nilradical homology by affine Weyl representatives. Under Proposition 3.3 this cohomology is the E_2 -page of the PBW spectral sequence. The diagonal support is the concentration input needed for Koszulness; any rank-by-rank Hilbert series for the chiral bar cohomology requires the additional Orlik–Solomon form factor and is recorded only where explicitly computed. $\square$

Corollary 4.8 ($\mathfrak{sl}_2$ specialization). *For $\mathfrak{g} = \mathfrak{sl}_2$ ($r = 1$), the verified chiral bar cohomology dimensions are*

$$\dim H^n(\bar{B}(V_k(\mathfrak{sl}_2))) = 2n + 1 \quad \text{for all } n \geq 1,$$

with rational generating function

$$P_{\widehat{\mathfrak{sl}_2}}(x) = \frac{x(3-x)}{(1-x)^2} = 3x + 5x^2 + 7x^3 + 9x^4 + \cdots.$$

Proof. The Chevalley–Eilenberg complex of $\mathfrak{sl}_2 \otimes t^{-1}\mathbb{C}[t^{-1}]$ is an exterior algebra on 3 generators per negative mode. The Garland–Lepowsky affine nilradical computation identifies the PBW spectral sequence page with the relevant $\mathfrak{sl}_2$ -invariant subalgebra, generated by one Casimir per mode t^{-m} , $m \geq 1$ (the quadratic Casimir $C_2^{(-m)} = e_{-m}f_{-m} + f_{-m}e_{-m} + h_{-m}^2/2$).

At bar degree n , the surviving cohomology lives at conformal weight $h = n(n+1)/2$ (the unique weight at which the n -fold exterior product of distinct modes concentrates). The dimension count proceeds: the n -th exterior power of an infinite sequence of $\mathfrak{sl}_2$ -adjoint modules, restricted to weight $n(n+1)/2$, has dimension $2n + 1$. This can be verified by the recurrence: at degree n , the new mode t^{-n} couples with all $n-1$ previous modes, and the representation-theoretic content grows by 2 at each step (from the tensor product rule for $\mathfrak{sl}_2$: $V_1 \otimes V_{n-1} = V_n \oplus V_{n-2}$, and the cohomological restriction selects V_n). $\square$

Proposition 4.9 (General rank bar cohomology at low degree). *For any simple $\mathfrak{g}$ of rank r :*

- (i) $H^0(\bar{B}(V_k(\mathfrak{g}))) = \mathbb{C}$ (the vacuum).
- (ii) $H^1(\bar{B}(V_k(\mathfrak{g}))) \cong \mathfrak{g}$ as a vector space (the generators), with $\dim H^1 = \dim \mathfrak{g}$.

(iii) At bar degree 2, the total dimension $\dim H^2(\bar{B}(V_k(\mathfrak{g})))$ depends on the full structure of the chiral bar complex and grows with the rank. For $\mathfrak{sl}_2$: $\dim H^2 = 5$. For $\mathfrak{sl}_3$: $\dim H^2 = 36$.

Proof. Part (i) is immediate: the degree-0 bar complex is $\mathbb{C}$, with $d_B|_{\bar{B}^0} = 0$.

Part (ii): at bar degree 1, the reduced bar complex is $s^{-1}\bar{V}_k(\mathfrak{g})$. Its weight-1 subspace is $s^{-1}\mathfrak{g}$. The reduced bar differential has no unary term on these indecomposable current generators, and no bar-degree-2 boundary has weight 1. Hence the surviving cohomology is $\mathfrak{g}$, the generator space of the Koszul dual.

Part (iii): at bar degree 2, the chain group is $B^2 = \mathfrak{g}^{\otimes 2} \otimes \Omega_{\log}^1$ of dimension $(\dim \mathfrak{g})^2$, and the bar differential $d_B^{(2)}: B^2 \rightarrow B^1$ extracts the OPE residue. The cohomology $H^2 = \ker d_B^{(2)} / \text{im } d_B^{(3)}$. For $\mathfrak{sl}_2$: $\dim B^2 = 9$, $\text{rk}(d_B^{(2)}) = 3$ (onto $B^1 = \mathfrak{g}$), $\text{rk}(d_B^{(3)}) = 1$, giving $\dim H^2 = 9 - 3 - 1 = 5$ (the spin-2 representation under the adjoint $\mathfrak{sl}_2$ -action, $\dim V_2 = 5$). For $\mathfrak{sl}_3$: $\dim B^2 = 64$, and the full computation gives $\dim H^2 = 36$ (see [9], Computation 7.2.1 and Remark 4.4). $\square$

Remark 4.10. The linear growth $\dim H^n = 2n + 1$ for $\mathfrak{sl}_2$ is exceptional among Koszul chiral algebras. Free fields (Heisenberg, free fermions) have partition-function growth $\dim H^n = p(n)$. Higher-rank Kac–Moody algebras, Virasoro, and $\mathcal{W}$ -algebras all have exponential growth. Among all standard Koszul algebras, $\widehat{\mathfrak{sl}}_2$ is the unique interacting algebra with sub-exponential bar cohomology growth.

Remark 4.11 (Growth rate hierarchy). The bar cohomology growth rate stratifies the standard landscape:

Family	Growth	Rate
Heisenberg	$p(n)$	sub-exponential
$\widehat{\mathfrak{sl}}_2$	$2n + 1$	linear
bc ghost	$2^n - n + 1$	exponential (2^n)
$\beta\gamma$	$\sim 3^n / \sqrt{n}$	exponential (3^n)
Virasoro	$M(n+1) - M(n)$	exponential (3^n)
$\widehat{\mathfrak{sl}}_3$	(conjectural)	exponential (8^n)

Here $p(n)$ is the partition function and $M(n)$ denotes Motzkin numbers. The $\widehat{\mathfrak{sl}}_3$ row records the expected exponential rate beyond the computed low-degree terms. The proved statement here is the class- L shadow-depth classification; the precise asymptotic growth for every rank $r \geq 2$ remains a separate bar-cohomology computation.

5 The Koszul dual of $V_k(\mathfrak{g})$

5.1 Level-shifting duality

Theorem 5.1. *For any simple $\mathfrak{g}$ and non-critical level $k \neq -h^\vee$, the Koszul dual of the universal affine vertex algebra is the chiral Chevalley–Eilenberg algebra at the Feigin–Frenkel dual level:*

$$(V_k(\mathfrak{g}))^\dagger \simeq \text{CE}^{\text{ch}}(V_{k'}(\mathfrak{g})), \quad k' = -k - 2h^\vee.$$

The Koszul dual has the same modular characteristic κ as $V_{k'}(\mathfrak{g})$ but is a different chiral algebra: it is the linear dual of the bar cohomology, not the vertex algebra $V_{k'}(\mathfrak{g})$ itself.

Proof. The bar differential (3.2) has two components: the double-pole residue $k (J^a, J^b) \cdot |o\rangle$ and the simple-pole residue $f^{ab}{}_c J^c$. Verdier duality on the Fulton–MacPherson compactification transforms the level- k $\mathcal{D}$ -module to the level- $(-k-2h^\vee)$ module:

$$\mathcal{L}_k^\vee \otimes \omega_{\overline{C}_n(X)} \cong \mathcal{L}_{-k-2h^\vee}.$$

The shift $2h^\vee$ arises from the adjoint Casimir: the dualizing sheaf $\omega_{\overline{C}_2}$ has a first-order pole along the diagonal whose residue is the Casimir eigenvalue $2h^\vee$ acting on $\mathfrak{g} \otimes \mathfrak{g}$ via the adjoint representation. This is the same $2h^\vee$ appearing in the Sugawara denominator $k + h^\vee$ and in the Killing form computation of Proposition 3.2.

The cobar OPE on the dual generators is therefore

$$J^{a,*}(z) J^{b,*}(w) \sim \frac{(-k-2h^\vee)(J^a, J^b)}{(z-w)^2} + \frac{f^{ab}{}_c J^{c,*}(w)}{z-w},$$

which is the affine OPE at level $k' = -k - 2h^\vee$. $\square$

Remark 5.2 (Involutivity). The level involution $k \mapsto k' = -k - 2h^\vee$ is involutive: $-(-k-2h^\vee) - 2h^\vee = k$. At the level of modular characteristics, $\kappa((V_k(\mathfrak{g}))^\dagger) = \kappa(V_{k'}(\mathfrak{g}))$. The critical level $k = -h^\vee$ is the unique fixed point of this involution.

Remark 5.3 (The Koszul dual is not the Feigin–Frenkel reflection). Three operations on affine vertex algebras share surface similarity but are mathematically distinct (see [9], §A.3):

- (1) *Koszul duality:* $V_k(\mathfrak{g}) \mapsto (V_k(\mathfrak{g}))^\dagger = \text{CE}^{\text{ch}}(V_{k'}(\mathfrak{g}))$. The output is the linear dual of bar cohomology, a different chiral algebra from $V_{k'}(\mathfrak{g})$.
- (2) *Feigin–Frenkel involution:* $k \mapsto -k - 2h^\vee$ within the same family. The output is $V_{k'}(\mathfrak{g})$, the universal vertex algebra at the reflected level.
- (3) *Negative-level substitution:* $V_k(\mathfrak{g}) \mapsto V_{-k}(\mathfrak{g})$, replacing the level parameter.

The modular characteristics of (1) and (2) coincide: both equal $\kappa(V_{k'}(\mathfrak{g}))$. But the algebras are different. The Koszul dual carries the structure of a coalgebra dual; the Feigin–Frenkel reflection preserves the algebra structure.

5.2 Modular characteristic

Proposition 5.4. *The modular characteristic of $V_k(\mathfrak{g})$ is*

$$\kappa(\widehat{\mathfrak{g}}_k) = \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee}, \quad (5.1)$$

and the complementarity sum vanishes: $\kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}_{k'}) = 0$.

Proof. The modular characteristic is the genus-1 obstruction class coefficient: $\text{obs}_1(V_k(\mathfrak{g})) = \kappa \cdot \lambda_1$, where λ_1 is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_{1,1}$. For the affine vertex algebra, the genus-1 partition function trace gives $F_1 = -\kappa \cdot \log \eta(\tau)$ where $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$. Comparing with the explicit computation $F_1(V_k(\mathfrak{g})) = -\frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee} \cdot \log \eta(\tau)$, we read off (5.1).

The complementarity sum is $\kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}_{k'}) = \frac{\dim \mathfrak{g}}{2h^\vee} [(k+h^\vee) + (-k-2h^\vee+h^\vee)] = \frac{\dim \mathfrak{g}}{2h^\vee} \cdot 0 = 0$. $\square$

Remark 5.5. The modular characteristic $\kappa(\widehat{\mathfrak{g}}_k)$ is distinct from the central charge $c(V_k(\mathfrak{g})) = k \dim \mathfrak{g} / (k + h^\vee)$. They coincide only for $\mathfrak{sl}_2$ at $k = 4$ and are generically unequal. The formula $\kappa = c/2$ applies to the Virasoro algebra, not to affine Kac–Moody algebras of rank > 1 .

6 Shadow depth

The shadow obstruction tower of a chirally Koszul algebra $\mathcal{A}$ is the sequence of invariants $(\kappa, \mathfrak{C}, \mathfrak{Q}, \dots)$ extracted as finite-order projections of the universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \text{MC}(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$: κ at degree 2 (the modular characteristic), the cubic shadow $\mathfrak{C}$ at degree 3, the quartic obstruction class $\mathfrak{Q}$ at degree 4, and so on. The shadow depth $r_{\max}(\mathcal{A})$ is the smallest degree at which the tower terminates (all higher obstructions vanish), or ∞ if it does not terminate.

Definition 6.1. The *shadow depth classification* partitions chirally Koszul algebras into four classes:

Class	$r_{\max}$	Name	Archetype
G	2	Gaussian	Heisenberg
L	3	Lie/tree	Affine Kac–Moody
C	4	Contact/quartic	$\beta\gamma$ system
M	∞	Mixed	Virasoro, $\mathcal{W}_N$

Theorem 6.2. Every affine Kac–Moody algebra $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$ is class L with shadow depth $r_{\max} = 3$. Concretely:

- (i) The degree-2 projection is the modular characteristic $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee) / (2h^\vee) \neq 0$.
- (ii) The degree-3 projection is the cubic shadow

$$\mathfrak{C}(x, y, z) = (x, [y, z]),$$

where $(\cdot, \cdot)$ is the invariant bilinear form on $\mathfrak{g}$ normalized so that the highest root satisfies $(\theta, \theta) = 2$, and $[\cdot, \cdot]$ is the Lie bracket. This is nonzero precisely because $\mathfrak{g}$ is nonabelian.

- (iii) The degree-4 quartic obstruction class vanishes: $o_4(\widehat{\mathfrak{g}}_k) = 0$. The vanishing is a consequence of the Jacobi identity for $\mathfrak{g}$.
- (iv) The higher obstruction classes vanish in the strict class-L representative supplied by the quadrichotomy theorem.

Proof. The degree-2 projection is computed in Proposition 5.4.

For the cubic shadow, the degree-3 component of $\Theta_{\widehat{\mathfrak{g}}_k}$ lives in the cyclic deformation complex $\text{Def}_{\text{cyc}}^{\text{mod}}$ at degree 3. The relevant graph sum reduces to a trivalent tree with three external legs; the amplitude evaluates to the structure constants f^{abc} contracted with the invariant form $(\cdot, \cdot)$, giving $\mathfrak{C}(x, y, z) = (x, [y, z])$. This vanishes if and only if $\mathfrak{g}$ is abelian.

The quartic obstruction o_4 lives in $H^2(F^4\mathfrak{g}^{\text{mod}}/F^5\mathfrak{g}^{\text{mod}}, d_2)$. The d_2 differential at degree 4 involves the graph sum over trivalent trees with 4 external legs and one internal edge. The obstruction is the class of $[\Theta_3, \Theta_3]$ in the Lie bracket of the convolution algebra. The Jacobi identity $f^{a[b}{}_e f^{c]d}{}_f + \text{cyclic} = 0$ forces this bracket to vanish in cohomology: the quadri-linear form $\sum_e f^{ab}{}_e f^{cd}{}_e$ satisfies the crossing symmetry imposed by the Jacobi identity, and the resulting cohomology class is trivially zero. The

vanishing of the remaining obstruction tower is the class- L strict-representative statement: after the quadrichotomy theorem has put the affine Kac–Moody MC element into Lie/tree normal form, no new independent higher shadow remains beyond the cubic Lie tensor. $\square$

Remark 6.3 (The collision residue). The trace-form collision residue (binary genus-0 shadow) of $V_k(\mathfrak{g})$ is

$$r^{\text{coll}}(z) = k \Omega_{\text{tr}}/z.$$

The KZ conformal-block connection uses the separate Sugawara-normalised kernel

$$r^{\text{KZ}}(z) = \Omega_{\text{KZ}}/((k + h^\vee)z).$$

At critical level the KZ coefficient is undefined, while the trace-form collision residue remains finite. The comparison between these two kernels is a normalization bridge, not equality as rational functions of k .

Remark 6.4 (The critical discriminant). The critical discriminant $\Delta = 8\kappa S_4 = 0$ vanishes for all affine Kac–Moody algebras (since $S_4 = 0$). The shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2$ is a perfect square, consistent with finite shadow depth. This is the algebraic characterization of class L : the shadow metric has a rational square root, and the shadow connection has monodromy -1 (the Koszul sign).

7 Examples

7.1 $\widehat{\mathfrak{sl}}_2$

The simplest nonabelian case. Set $\mathfrak{g} = \mathfrak{sl}_2$, $\dim \mathfrak{g} = 3$, $h^\vee = 2$, $\text{rk} = 1$, $|\Delta_+| = 1$.

Computation 7.1. (i) *Koszul dual level*: $k' = -k - 4$.

(ii) *Modular characteristic*: $\kappa(\widehat{\mathfrak{sl}}_{2,k}) = 3(k+2)/4$.

(iii) *Central charge*: $c(\widehat{\mathfrak{sl}}_{2,k}) = 3k/(k+2)$.

(iv) *Complementarity*: $\kappa + \kappa' = 3(k+2)/4 + 3(-k-4+2)/4 = 3(k+2)/4 + 3(-k-2)/4 = 0$.

(v) *Bar cohomology*: $\dim H^n = 2n + 1$ for $n \geq 1$ (Corollary 4.8). Generating function $x(3-x)/(1-x)^2$.

(vi) *r-matrix*: the trace-form collision residue is $r^{\text{coll}}(z) = k \Omega_{\text{tr}}/z$. The KZ kernel is $r^{\text{KZ}}(z) = \Omega_{\text{KZ}}/((k+2)z)$.

(vii) *Shadow class*: L , $r_{\max} = 3$. Cubic shadow $\mathfrak{C}(x, y, z) = (x, [y, z])$ (the Killing form of x with $[y, z]$); quartic vanishes by $\mathfrak{sl}_2$ Jacobi.

Remark 7.2. The simple quotient $L_k(\mathfrak{sl}_2)$ at admissible levels $k = -2 + p/q$ (with $p \geq 2$, $q \geq 1$, $\gcd(p, q) = 1$) is also chirally Koszul. The argument uses the structural fact that $L_k(\mathfrak{sl}_2)$ has a single family of null vectors at each admissible level, and the bar cohomology is controlled by the Kac–Wakimoto character formula.

7.2 $\widehat{\mathfrak{sl}}_3$

Set $\mathfrak{g} = \mathfrak{sl}_3$, $\dim \mathfrak{g} = 8$, $h^\vee = 3$, $\text{rk} = 2$, $|\Delta_+| = 3$. The rank-2 case is the first where a second independent Casimir appears (the cubic Casimir $C_3 = d_{abc} J^a J^b J^c$), and the Koszul dual structure acquires genuinely new features relative to $\mathfrak{sl}_2$.

Computation 7.3. (i) *Koszul dual level*: $k' = -k - 6$.

- (ii) *Modular characteristic*: $\kappa(\widehat{\mathfrak{sl}}_{3,k}) = 4(k+3)/3$. Verification: $\dim(\mathfrak{sl}_3) = 8$, $h^\vee = 3$, so $\kappa = 8(k+3)/(2 \cdot 3) = 4(k+3)/3$.
- (iii) *Central charge*: $c(\widehat{\mathfrak{sl}}_{3,k}) = 8k/(k+3)$.
- (iv) *Complementarity*: $\kappa(\widehat{\mathfrak{sl}}_{3,k}) + \kappa(\widehat{\mathfrak{sl}}_{3,k'}) = \frac{4(k+3)}{3} + \frac{4(-k-6+3)}{3} = \frac{4(k+3)+4(-k-3)}{3} = 0$.
- (v) *Bar cohomology at low degree*: $\dim H^1 = 8$ (the adjoint representation). The total degree-2 bar cohomology has $\dim H^2 = 36$, computed by direct enumeration on the chiral bar complex $B^2 = \mathfrak{g}^{\otimes 2} \otimes \Omega_{\log}^1$ (see [9], Computation 7.2.1). This differs from the Chevalley–Eilenberg value $\dim H_{\text{CE}}^2(\mathfrak{n}^{\text{loop}}_-) = 20$ because the chiral bar complex carries the Orlik–Solomon form factor, which is absent from the exterior algebra $\Lambda^*(\mathfrak{n}^{\text{loop}}_-)^*$. The computed initial segment of the generating function is

$$P_{\widehat{\mathfrak{sl}}_3}(x) = 8x + 36x^2 + O(x^3).$$

A closed form and the full asymptotic remain conjectural.

- (vi) *Shadow class*: L , $r_{\max} = 3$. The cubic Lie/tree shadow is $\mathfrak{C}(x, y, z) = (x, [y, z])$; it is governed by the structure constants f^{abc} .
- (vii) *r-matrix*: $r^{\text{coll}}(z) = k \Omega_{\text{tr}}/z$. The KZ kernel is $r^{\text{KZ}}(z) = \Omega_{\text{KZ}}/((k+3)z)$.
- (viii) *Koszul conductor*: $K = 2 \cdot 8 = 16$.

Remark 7.4. The $\mathfrak{sl}_3$ case exhibits two qualitative differences from $\mathfrak{sl}_2$. First, the ring of $\mathfrak{sl}_3$ -invariants in the symmetric algebra on the adjoint is generated by Casimirs of degrees 2 and 3 (rather than a single quadratic Casimir for $\mathfrak{sl}_2$), producing faster-growing bar cohomology. Second, the cubic Casimir d_{abc} provides a new channel for the cubic shadow $\mathfrak{C}$ that does not exist for $\mathfrak{sl}_2$ (where all cubic invariants are proportional to the structure constants). The shadow class remains L because the quartic obstruction still vanishes by the Jacobi identity, which holds for all simple Lie algebras regardless of rank.

7.3 Exceptional types

Proposition 7.5. *For the simply-laced exceptional Lie algebras $\mathfrak{g} \in \{E_6, E_7, E_8\}$, the invariants of $V_k(\mathfrak{g})$ are:*

$\mathfrak{g}$	$\dim$	$h^\vee$	$\kappa(\widehat{\mathfrak{g}}_k)$	K	<i>Class</i>
E_6	78	12	$13(k+12)/4$	156	L
E_7	133	18	$133(k+18)/36$	266	L
E_8	248	30	$62(k+30)/15$	496	L

Here $K = 2 \dim \mathfrak{g}$ is the Koszul conductor. Every entry satisfies $\kappa + \kappa' = 0$ and has shadow depth $r_{\max} = 3$ (class L).

Proof. The formula $\kappa = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$ is universal for all simple $\mathfrak{g}$ (Proposition 5.4). The Koszul conductor is $K = c + c'$; since $c(V_k(\mathfrak{g})) = k \dim \mathfrak{g}/(k+h^\vee)$ and $c(V_{k'}(\mathfrak{g})) = (-k-2h^\vee) \dim \mathfrak{g}/(-k-h^\vee)$, we compute $K = \dim \mathfrak{g}[k/(k+h^\vee) + (-k-2h^\vee)/(-k-h^\vee)]$. Simplifying:

$$\frac{k}{k+h^\vee} + \frac{-k-2h^\vee}{-k-h^\vee} = \frac{k}{k+h^\vee} + \frac{k+2h^\vee}{k+h^\vee} = \frac{2(k+h^\vee)}{k+h^\vee} = 2.$$

Therefore $K = 2 \dim \mathfrak{g}$, universal for all simple $\mathfrak{g}$. Shadow class L follows from Theorem 6.2. $\square$

Remark 7.6 (Non-simply-laced types). For non-simply-laced $\mathfrak{g} \in \{B_n, C_n, G_2, F_4\}$, the lacing number $r^\vee = |\alpha_{\text{long}}|^2/|\alpha_{\text{short}}|^2$ enters the formulas only through the dual Coxeter number $h^\vee$ and dimension. The shadow depth remains $r_{\text{max}} = 3$ (class L), the Koszul conductor is still $K = 2 \dim \mathfrak{g}$, and $\kappa + \kappa' = 0$. All non-simply-laced affine Kac–Moody algebras are chirally Koszul with the same structural properties as the simply-laced cases. Specific values: $\kappa(\widehat{G}_{2,k}) = 7(k+4)/4$, $\kappa(\widehat{F}_{4,k}) = 26(k+9)/9$.

8 The critical level

The critical level $k = -h^\vee$ is the fixed point of the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ and a singular point for both the Sugawara construction and the modular characteristic. The bar complex simplifies qualitatively.

Proposition 8.1. *At the critical level $k = -h^\vee$:*

- (i) *The curvature vanishes: $m_1^2 = 0$ (Proposition 3.2 with $k + h^\vee = 0$). The bar complex is a strict dg coalgebra (uncurved).*
- (ii) *The modular characteristic vanishes: $\kappa(\widehat{\mathfrak{g}}_{-h^\vee}) = 0$. The genus-1 obstruction class $\text{obs}_1 = 0$, and the partition function $F_1 = 0$.*
- (iii) *The Sugawara construction is undefined: the denominator $k + h^\vee = 0$ in the Sugawara formula $L_{\text{Sug}} = \frac{1}{2(k+h^\vee)} \sum_a J^a J^a$ is singular. No Virasoro subalgebra exists.*
- (iv) *The center of the completed enveloping algebra enlarges to the Feigin–Frenkel center $\mathfrak{z}(\widehat{\mathfrak{g}})$ ([3]), which is isomorphic to the algebra of functions on the space of $\mathfrak{g}^\vee$ -opers on the formal disc.*
- (v) *Under the local completed bar–Ext comparison, the critical completed bar–Ext groups identify with de Rham forms on the local oper space:*

$$H_{\text{bExt}}^n \cong \Omega^n(\text{Op}_{\mathfrak{g}^\vee}(D)).$$

This is not a statement about raw bar cohomology.

Proof. Items (i)–(ii) follow directly from Proposition 3.2 and equation (5.1) at $k = -h^\vee$. Item (iii) is classical [3]. Item (iv) is the Feigin–Frenkel theorem [3]. Item (v) is the completed bar–Ext comparison combined with the fundamental local equivalence at the critical level, which identifies the relevant completed critical category with the Whittaker category over $\text{Op}_{\mathfrak{g}^\vee}(D)$. $\square$

Remark 8.2 (Critical level and level reflection). The critical level is the unique self-dual point of the Verdier level reflection: $(-h^\vee)' = -(-h^\vee) - 2h^\vee = -h^\vee$. Full Koszulness fails at this point; self-duality is only the scalar level reflection, not an identification of the raw critical vertex algebra with its strict Koszul dual. The vanishing of κ at the critical level is consistent with the self-duality: $\kappa + \kappa' = 2\kappa = 0$.

In the shadow obstruction tower, $\kappa = 0$ means the degree-2 projection vanishes. However, the cubic shadow $\mathfrak{C}$ can still be nonzero (the Lie bracket does not vanish at the critical level), so the critical-level algebra is not class G (Gaussian). The classification is: at $k = -h^\vee$, the algebra has $\kappa = 0$ with $\mathfrak{C} \neq 0$, a degenerate class- L configuration where the leading scalar invariant vanishes but the cubic structure persists.

Remark 8.3 (The geometric Langlands connection). The critical-level completed bar–Ext complex lies on the oper side of local geometric Langlands. Feigin–Frenkel identifies

$$\mathfrak{z}(\widehat{\mathfrak{g}}) \cong \text{Fun}(\text{Op}_{\mathfrak{g}^\vee}(D)).$$

Together with the completed bar–Ext comparison and Raskin’s local equivalence, this identifies critical-level completed bar–Ext groups with de Rham forms on the local oper space. The global curve X replaces the formal disc by the Ran-space factorisation algebra; this is the local input, not by itself the global Langlands kernel.

9 The affine Kac–Moody point in the Koszul landscape

The affine Kac–Moody calculation occupies the class- L vertex of the modular Koszul landscape. The PBW universality criterion identifies $V_k(\mathfrak{g})$ as chirally Koszul because it is freely strongly generated and its associated graded is classically Koszul. Verdier duality gives the Koszul dual

$$(V_k(\mathfrak{g}))^\dagger \simeq \text{CE}^{\text{ch}}(V_{-k-2h^\vee}(\mathfrak{g})).$$

The genus-one obstruction coefficient is

$$\kappa(\widehat{\mathfrak{g}}_k) = \frac{\dim \mathfrak{g} (k + h^\vee)}{2h^\vee}, \quad \kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}_{-k-2h^\vee}) = 0.$$

The cubic shadow is the invariant form $(x, [y, z])$, and Jacobi annihilates the quartic obstruction; hence the shadow depth is $r_{\max} = 3$.

- (1) Chiral Koszulness of $V_k(\mathfrak{g})$ (Theorem 4.2) is a special case of the PBW universality criterion (Proposition 5.3.1 of [9]): every freely strongly generated chiral algebra whose associated graded is classically Koszul is chirally Koszul. This criterion also applies to universal $\mathcal{W}$ -algebras, lattice vertex algebras, and the $\beta\gamma$ system.
- (2) The Koszul dual identification (Theorem 5.1) is a special case of the bar-cobar adjunction (Theorem A of [9]) combined with the Verdier intertwining theorem.
- (3) The modular characteristic formula (Proposition 5.4) and the complementarity sum $\kappa + \kappa' = 0$ are special cases of Theorem D of [9]. The vanishing complementarity sum is specific to the Kac–Moody and free-field families. For $\mathcal{W}$ -algebras the scalar complementarity is nonzero and is computed by the family-specific conductor normalization, not by a universal affine ratio.
- (4) The shadow depth classification (Theorem 6.2) places affine Kac–Moody algebras in class L within the four-class partition $G/L/C/M$ of [9]. The four classes are: Gaussian (Heisenberg, $r_{\max} = 2$), Lie/tree (affine KM, $r_{\max} = 3$), contact/quartic ($\beta\gamma$, $r_{\max} = 4$), and mixed (Virasoro, $\mathcal{W}_N$, $r_{\max} = \infty$). All four classes are chirally Koszul; shadow depth classifies complexity within the Koszul world, not Koszulness itself.
- (5) The trace-form collision residue $r^{\text{coll}}(z) = k\Omega_{\text{tr}}/z$ (Remark 6.3) is the binary genus-0 projection of the universal Maurer–Cartan element $\Theta_{\widehat{\mathfrak{g}}_k}$. The KZ coefficient $\Omega_{\text{KZ}}/((k + h^\vee)z)$ belongs to the conformal-block normalization. The affine modular characteristic is recovered from the double-pole channel together with the Sugawara shift:

$$\kappa_{\text{dp}} + \dim(\mathfrak{g})/2 = \kappa(\widehat{\mathfrak{g}}_k).$$

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